

OsdagBridge LaTeX Report Generator Refactoring and Enhancements

Technical Report

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Contents

1. Problem in the existing generator	3
2. Approach	3
3. Repeated headers (longtable retrofit)	3
4. Footer overlap and page geometry	4
5. Chapter 3 live-load tables	5
5.1 Vehicle Live Loads	5
5.2 Centrifugal Force	5
5.3 Footpath Live Loads	5
6. Utilization Ratio charts	6
7. Material quantity charts	7
8. styles.py as the layout source	9
9. Files changed	10
10. Validation	10
11. Conclusion	10

1. Problem in the existing generator

OsdagBridge writes a multi-chapter design PDF through string-built LaTeX in `core/reports/`. The compile path is PyLaTeX / pdflatex. The baseline output (saved here as `Report_Before.pdf`, 42 pages) already had content, but four classes of defect made it hard to use as a design document.

Defect	What actually happened in the PDF
Multi-page tables	Chapter tables used <code>longtable</code> without a filled <code>\endhead</code> block (Chapter 2 restraint table was empty between <code>\endfirsthead</code> and <code>\endhead</code>). After a page break the reader saw data rows with no column titles.
Footer collision	Bottom margin was tight and footer/figure <code>\vspace</code> was negative. On dense Chapter 5 pages a girder label such as G4 sat on the green footrule.
Live loads	Vehicle LL and footway load sat in one poorly structured block. Impact factor, braking, eccentricity and centrifugal force were not readable as columns, and rows did not follow Loading-tab checkboxes.
No UR / BOQ plots	Table 5.22 listed demand/capacity numbers only. Chapter 7 listed tonnage in a table with no chart. Pass/fail against UR = 1.0 was not visible at a glance.
Scattered style	Margins, <code>\tabcolsep</code> , brand green and footer clearance were hard-coded in the preamble and in individual chapters.

The work stays inside the existing string-templated architecture. New chapters were not rewritten as PyLaTeX Document objects; helpers emit the same `longtable` dialect the compiler already accepts, then a final pass rewrites every table in the assembled `.tex`.

2. Approach

- One helper for repeating headers; one retrofit pass so every chapter `longtable` keeps its header after a page break.
- Geometry and paddings live in `styles.py`; the preamble reads that module.
- Chapter 3 live loads are built from Loading-tab keys (`KEY_LL_IRC_*`).
- Charts are written as PNG under the report `assets/` folder during compile and embedded with `\includegraphics`. Matplotlib (Agg) is preferred; Pillow is used when Agg is unavailable.
- `Report_After.pdf` is produced by the same `generate_report()` entry used by Generate Report in the desktop Output Dock.

3. Repeated headers (longtable retrofit)

`longtable_repeating_headers()` in `report_utils.py` writes the standard four-part header: first-page caption + header (`\endfirsthead`), continued caption + the same header (`\endhead`), a right-aligned “Continued on next page” foot, and `\endlastfoot` so the note is omitted on the last fragment.

```
\endfirsthead
\caption[]{\dots (continued)}\
<same header row(s)>
\endhead
\multicolumn{n}{r}{\footnotesize\textit{Continued on next page}}\
\endfoot \endlastfoot
```

New tables (Chapter 3 LL, Chapter 7 BOQ, Overall Design Check Summary) call this helper directly. Older chapter templates still emit a raw `\begin{longtable}...\end{longtable}`. Those are not edited one-by-one. After the full `.tex` is concatenated, `generate_report()` runs:

```
from osdagbridge.core.reports.report_utils import retrofit_longtables
full_tex = retrofit_longtables(full_tex)
```

`retrofit_longtables` walks every longtable. If `\endfirsthead` is already present it only inserts a missing `\endfoot`. Otherwise it splits caption / bold header / body and rebuilds the table through the helper. Column count is inferred from the `colspec` (l,c,r,L,C,R,X,m) ignoring nested braces, which is enough for the existing chapter specs.

In `Report_After.pdf` the continued-page marker appears on the long Chapter 5 stiffener and DCR tables (pages 8, 12, 19–31, 33–34, 36, 39, 44). `Report_Before.pdf` has almost no such markers (two `\endfirsthead` sites in the `.tex`, one of them empty).

Table 2.6: Girder General Information

Girder	Member ID	Design Mode	Girder Type	Girder Symmetry
G1	G1M1	Optimized	Welded I	Symmetric
G2	G2M1	Optimized	Welded I	Symmetric
G3	G3M1	Optimized	Welded I	Symmetric
G4	G4M1	Optimized	Welded I	Symmetric

Table 2.7: Girder Section Dimensions

Girder	Total Depth, D (mm)	Web, t_w (mm)	Top Flange (b_{tf} , t_{tf}) mm	Bottom Flange (b_{bf} , t_{bf}) mm
G1	1500 mm	12 mm	400 mm, 25 mm	450 mm, 32 mm
G2	1500 mm	12 mm	400 mm, 25 mm	450 mm, 32 mm
G3	1500 mm	12 mm	400 mm, 25 mm	450 mm, 32 mm
G4	1500 mm	12 mm	400 mm, 25 mm	450 mm, 32 mm

Continued on next page

Figure 1. `Report_After.pdf`— table fragment with repeating header and “Continued on next page”.

4. Footer overlap and page geometry

The reported failure was text (e.g. girder id G4) drawing into the footer box. Three changes, all numeric values from `styles.py`:

Setting	Value	Why
PAGE_MARGIN_BOTTOM	1.25in	Was 1in; extra room above the footer
FOOTER_CLEARANCE	6pt	Positive skip above the green rule; negative <code>vspace</code> removed
NEEDSPACE_BASELINES	8	<code>\BeforeBeginEnvironment{longtable}{\needspace{8\baselineskip}}</code>
LT_PRE / LT_POST	2pt / 8pt	Stops longtables sitting flush against the footer

`_fig_embed` no longer applies a negative `\vspace` after figures. On the After PDF, a G4 block that starts at the page bottom is followed by the continued note and then the Osdag | FOSSEE | IIT

Bombay footer, not overlapping it.

5. Chapter 3 live-load tables

`live_load_tables.py` is called from `chap3.py`. It emits three longtables so the published vehicle-LL example stays readable on A4.

5.1 Vehicle Live Loads

Columns match the task example: Vehicle; Total Load (kN); Impact Factor; Braking Load (Considered? / Value / Eccentricity). A row is emitted only if the matching Loading-tab key is selected. A single Class 70R variant is labelled Class 70R; multiple 70R variants keep Wheeled / Tracked / Bogie suffixes. Numeric cells always print a value or an em-dash.

UI vehicle	Input key
Class A	KEY_LL_IRC_CLASS_A
Class AA Wheeled / Tracked	KEY_LL_IRC_AA_WHEELED / AA_TRACKED
Class 70R Wheeled / Tracked / Bogie	KEY_LL_IRC_70R_*
Class SV	KEY_LL_IRC_CLASS_SV (braking: KEY_BL_IRC_CLASS_SV)
Class Fatigue	KEY_LL_IRC_CLASS_FATIGUE
Custom list	KEY_LL_CUSTOM_VEHICLES

Total load comes from the IRC helper and is converted N → kN. Impact factor uses Cl. 204 / Cl. 208 with span. Braking uses Cl. 211.2; eccentricity is KEY_LL_ECCENTRICITY or Cl. 211.3 default 1.2 m.

5.2 Centrifugal Force

A second table carries centrifugal force with unit columns: Vehicle; Considered?; Value (kN); Radius (m); Speed (km/h). Force follows IRC 6 Cl. 212, $F = W v^2 / (127 R)$. If no curve radius is set, Considered is No. The demo case uses R = 250 m, V = 80 km/h (Class A F = 109.55 kN; Class SV F = 761.32 kN).

5.3 Footpath Live Loads

Separate table: Parameter / Value / Unit / Reference. Rows include configuration (Both Sides / ...), applicability, pressure mode (IRC 6 Cl. 206.1 or user value), and intensity in kN/m². This is no longer merged into the vehicle table.

3 Loads and Load Combinations

This section summarizes all loads applied to the bridge and the load combinations considered for analysis and design.

Table 3.1: Dead Load – Self Weight

parameter	value
Steel Self-Weight Applied	78.5 kN/m ³
Concrete Deck Weight	25 kN/m ³
Self-Weight Factor	1.05

Table 3.2: Dead Load for Surfacing (DW)

parameter	value
Wearing Course Load	Bituminous x 65
Additional SIDL (Crash Barrier)	5.0 kN/m per barrier
Railing Load	0.75 kN/m*

Table 3.3: Vehicle Live Loads (LL)

Vehicle	Total Load (kN)	Impact Factor	Braking Load		
			Considered?	Value	Eccentricity
Class A	543.47	1.207	Yes	108.69	1.20
Class 70R	981.00	1.207	Yes	108.69	1.20
Class SV	3776.85	1.207	No	—	—

Note: Vehicle rows follow Additional Inputs → Loading tab selections. Total load and impact factor (1 + IM) per IRC 6 Cl. 204 / Cl. 208. Braking force is the carriageway value per Cl. 211.2 (shown on each vehicle for which braking is considered). Eccentricity is measured from the deck top per Cl. 211.3.

* Software default value

Figure 2. Report_After.pdf, Chapter 3 — Vehicle Live Loads (task example columns).

6. Utilization Ratio charts

After the Overall Design Check Summary table (Section 5.5 / Table 5.22 in the brief), `chap5.py` builds the four primary-element UR values and writes `assets/ur_summary.png`.

Bar	Source of UR
Steel Plate Girders	max DCR over girder flexure / shear / LTB / interaction rows
Concrete Deck Slab	max of sag/hog/overhang Mu/Md, punching, shear, crack checks
Cross Bracing	max of compression, tension, slenderness efficiencies
End Diaphragms	governing DCR for the end-diaphragm family

Y-axis is UR (Demand/Capacity). Bars are green if $UR \leq 1.0$ and red if $UR > 1.0$. A dashed line at 1.0 uses `styles.COLOR_UR_THRESHOLD / UR_THRESHOLD`. In the demo run: girders 0.82, deck 0.68, bracing 0.57, end diaphragms 0.54.

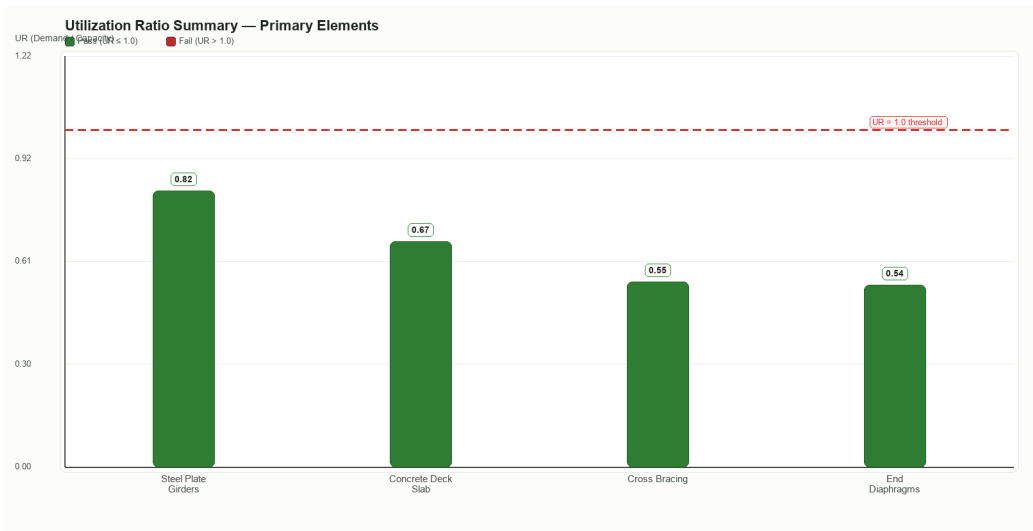


Figure 3. UR summary generated by report_charts.py (same figure embedded in Chapter 5).

Table 5.29: Overall Design Check Summary — All Members (continued)

Member / Check	Governing Load Combo	Demand	Capacity	UR
End Diaphragm — Shear	Basic ULS: 1.35DL + 1.5LL	120.40 kN	280.000 kN	0.43

Note: $UR = Demand / Capacity$. All values ≤ 1.0 indicate passing checks. The governing check for each component is highlighted in the individual design check sections above.

Utilization Ratio Visualisation

Governing demand/capacity ratios for the primary element families are plotted below. Bars are green when $UR \leq 1.0$ (pass) and red when $UR > 1.0$ (fail).

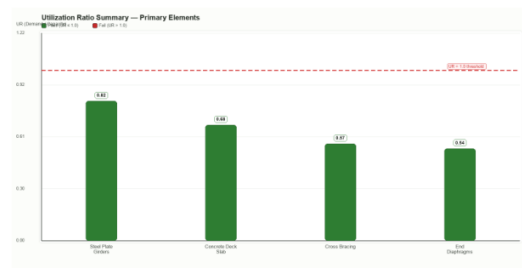


Figure — Utilization Ratio (Demand/Capacity) for Steel Plate Girders, Concrete Deck Slab, Cross Bracing and End Diaphragms. Red dashed line marks $UR = 1.0$ (pass/fail threshold).

Figure 4. Placement in Report_After.pdf, after the overall DCR table.

7. Material quantity charts

chap7.py still prints Table 7.1 (BOQ). After the table it calls save_steel_tonnage_chart and save_concrete_vs_rebar_chart. PNGs go to assets/steel_tonnage.png and assets/concrete_rebar.png.

Steel bars: Girders, CB Top, CB Bottom, CB Diagonal, End Diaphragms (ed_wt_total).
Concrete volume (m³) and reinforcement steel (MT) use separate panels with independent axes.

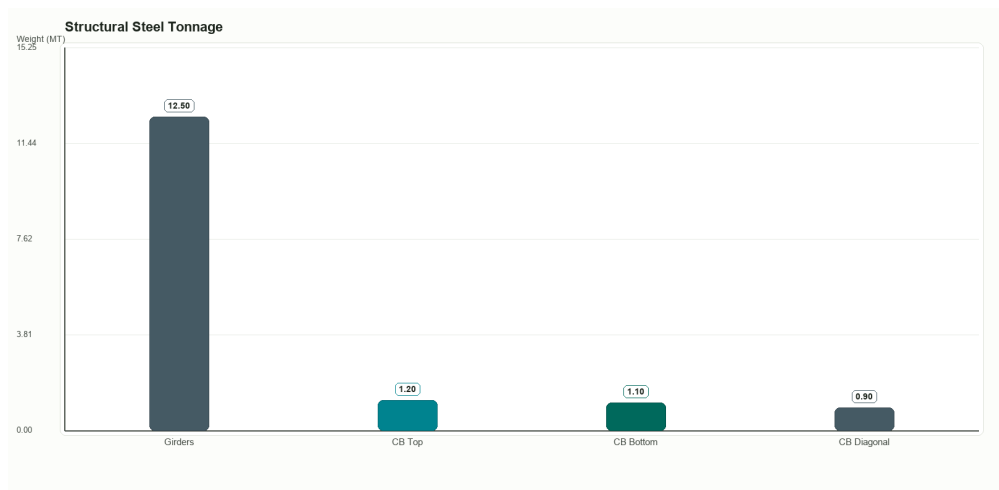


Figure 5. Structural steel tonnage (MT).

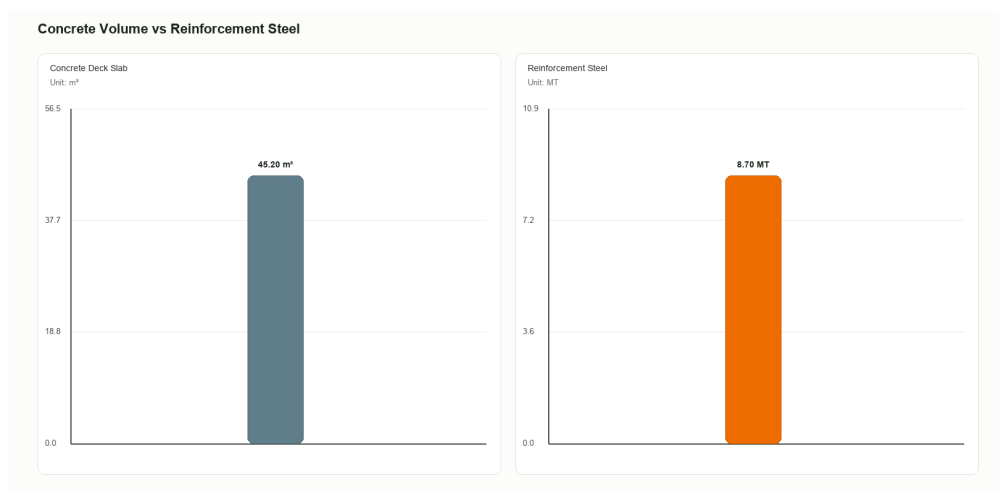


Figure 6. Concrete volume vs reinforcement steel.

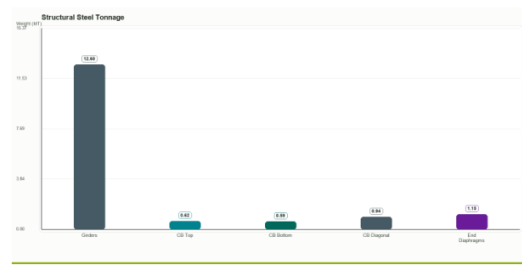


Figure — Structural steel tonnage (girders, cross bracing, end diaphragms).

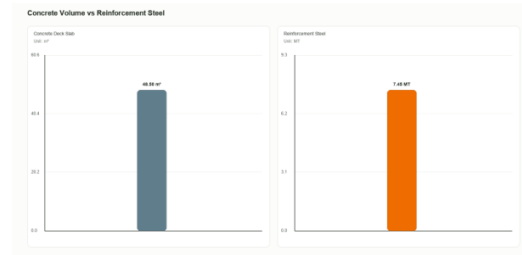


Figure — Concrete volume (m³) and reinforcement steel (MT) shown on independent axes.

Figure 7. The same figures inside Chapter 7 of Report_After.pdf.

8. styles.py as the layout source

Chapter generators import `osdagbridge.core.reports.styles`. The preamble calls `styles.geometry_options()` and `styles.table_layout_preamble()`. Bottom margin, brand green and table padding are changed in one place.

Group	Symbols
Geometry	PAGE_PAPER, PAGE_MARGIN_TOP/BOTTOM/LEFT/RIGHT
Colour	COLOR_OSDAG_GREEN, COLOR_UR_PASS/FAIL/THRESHOLD, chart colours
Tables	TABCOLSEP, ARRAYSTRETCH, EXTRAROWHEIGHT, LT_PRE/POST, ROW_SKIP, NEEDSPACE_BASELINES
Dense tables	TABCOLSEP_DENSE, ARRAYSTRETCH_DENSE, dense_table_begingroup/endgroup
Header/footer	HEADDRULE_HEIGHT, FOOTRULE_HEIGHT, FOOTER_CLEARANCE
Figures	FIGURE_WIDTH_DEFAULT, CHART_DPI, CHART_FIGSIZE_*, UR_THRESHOLD

Consumers: `report_generator.py` (preamble), `report_utils.py` (figure vspace, longtable foot), `report_charts.py` (bar colours, DPI, threshold), `chap2.py`–`chap5.py`, `chap7.py`, `live_load_tables.py` (row skip).

9. Files changed

Path under core/reports/	Role
styles.py	Added. Theme and geometry.
report_charts.py	Added. UR + steel + concrete/rebar PNGs.
live_load_tables.py	Added. Vehicle example table, centrifugal table, footpath table.
report_utils.py	longtable_repeating_headers, retrofit_longtables, _fig_embed spacing.
report_generator.py	styles preamble; assets_dir; retrofit before pdflatex.
chap2.py	Restraint-table header gap; imports styles.
chap3.py	Vehicle / centrifugal / footpath tables; ROW_SKIP from styles.
chap8.py	Code-table spacing from styles.py.
chap4.py	Reactions from analysis_summary['reactions']; styles import.
chap5.py	UR summary dict + chart after overall DCR table.
chap7.py	BOQ longtable + steel/concrete charts including ED tonnage.

10. Validation

Report_After.pdf was compiled with `generate_report()`, the same entry used by Generate Report in the Output Dock. The demo payload selects Class A, Class 70R and Class SV on the Loading tab, with curve radius 250 m. Vehicle LL appears on p.11; centrifugal and footpath on p.12; the UR figure on p.44; material figures on p.46; continued headers throughout Chapter 5. Report_Before.pdf (42 pages) is the baseline without those tables and figures.

Desktop call chain:

```
OutputDock._on_report_clicked()
→ PlateGirderBridge.generate_design_report()
→ build_report_payload(...)
→ generate_report(payload, request)
```

11. Conclusion

The six items in the brief are in `core/reports/` and in `Report_After.pdf`: repeating longtable headers, footer clearance, Loading-tab live-load tables, UR bars with a 1.0 threshold, Chapter 7 quantity charts, and `styles.py`. `Report_Before.pdf` is the control. `Technical_Changes.md` lists the files touched.